



Hands-On Activity: Combine multiple pieces of data



Activity Overview

In previous activities, you gained experience using spreadsheet functions for manipulating and cleaning data. In this activity, you'll use the `CONCAT` and `CONCATENATE` functions to help you quickly and efficiently combine multiple pieces of raw data into new data.

By the time you complete this activity, you will be able to use these functions to combine data. This will enable you to simplify and condense data, which is important for processing and cleaning data in your career as a data analyst.

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

To get started, you will need the **CONCAT** function exercise spreadsheet.

To use the template for the spreadsheet, click the link below and select “Use Template.”

Link to template: [CONCAT Function Exercise Spreadsheet](#)

OR

If you don't have a Google account, you can download the spreadsheet directly from the attachment below.



Occasionally, you will encounter a dataset with data values in separate cells that you want to combine as a single value in a single cell. This is common when dealing with names and dates. The dataset may have separate columns for first names and last names, but you may want a column with the full names.

City / state and month / year combinations are also often desirable to have together, as they are likely to be recorded together.

The **CONCAT** function in spreadsheets can combine these kinds of data.

First, using the spreadsheet you downloaded, you'll combine the two sets of names in columns **First Name** and **Last Name** in a new column called **Full Name**.

To do this, follow these steps:

1. Click on cell **F2**. This is where you start the data for the new column. After you click on the cell, enter **=CONCAT (A2 , B2)** into the function bar and hit **Enter (Windows)** or **Return (Mac)**.

fx =CONCAT(A2,B2)							
	A	B	C	D	E	G	H
1	First	Last	Month	Day	Year		
2	George	Washington	April	30	1789		
3	John	Adams	March	4	1797		
4	Thomas	Jefferson	March	4	1801		
5	James	Madison	March	4	1809		
6	James	Monroe	March	4	1817		
7	John Quincy	Adams	March	4	1825		
8	Andrew	Jackson	March	4	1829		
9	Martin	Van Buren	March	4	1837		
10	William Henry	Harrison	March	4	1841		
11	John	Tyler	April	4	1841		
12	James K.	Polk	March	4	1845		
13	Zachary	Taylor	March	4	1849		
14	Millard	Fillmore	July	9	1850		
15	Franklin	Pierce	March	4	1853		

Once you press enter, the following data should appear in the cell:

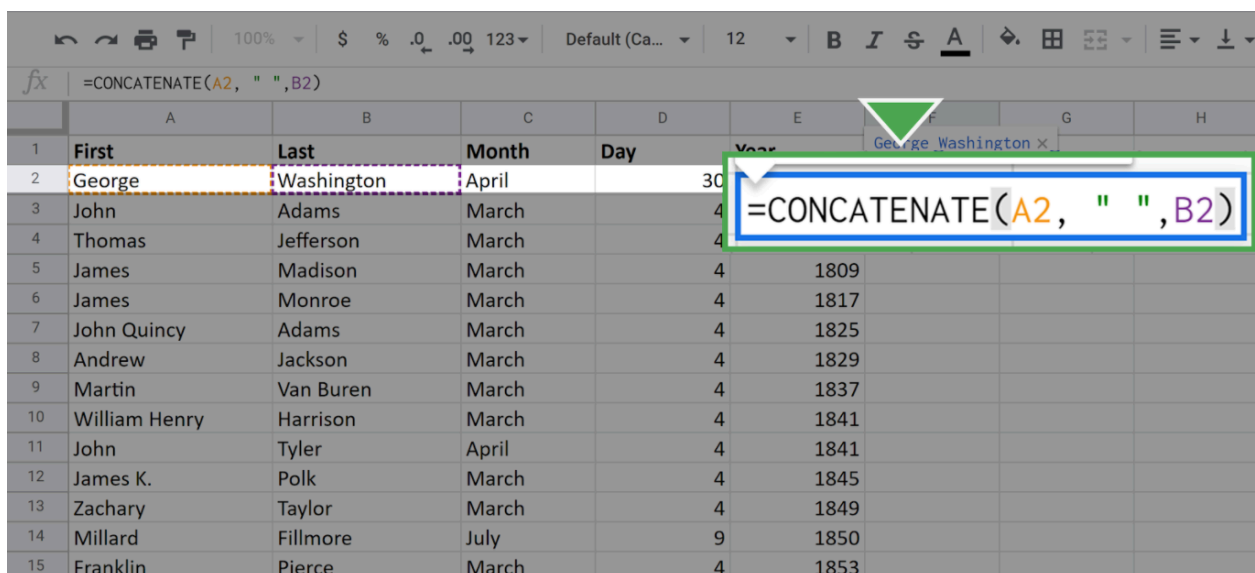
	A	B	C	D	E	F	G
	First	Last	Month	Day	Year		
	George	Washington	April	30	1789	GeorgeWashington	
	John	Adams	March	4	1797		
	Thomas	Jefferson	March	4	1801		
	James	Madison	March	4	1809		
	James	Monroe	March	4	1817		
	John Quincy	Adams	March	4	1825		
	Andrew	Jackson	March	4	1829		
	Martin	Van Buren	March	4	1837		
	William Henry	Harrison	March	4	1841		
	John	Tyler	April	4	1841		
	James K.	Polk	March	4	1845		
	Zachary	Taylor	March	4	1849		
	Millard	Fillmore	July	9	1850		
	Franklin	Pierce	March	4	1853		

You have merged or, technically, **concatenated** the two data values from cells **A2** and **B2**. Because you listed A2 first in the **CONCAT** function argument, it comes first in the final result.

Notice that the two names were combined without a space between them.

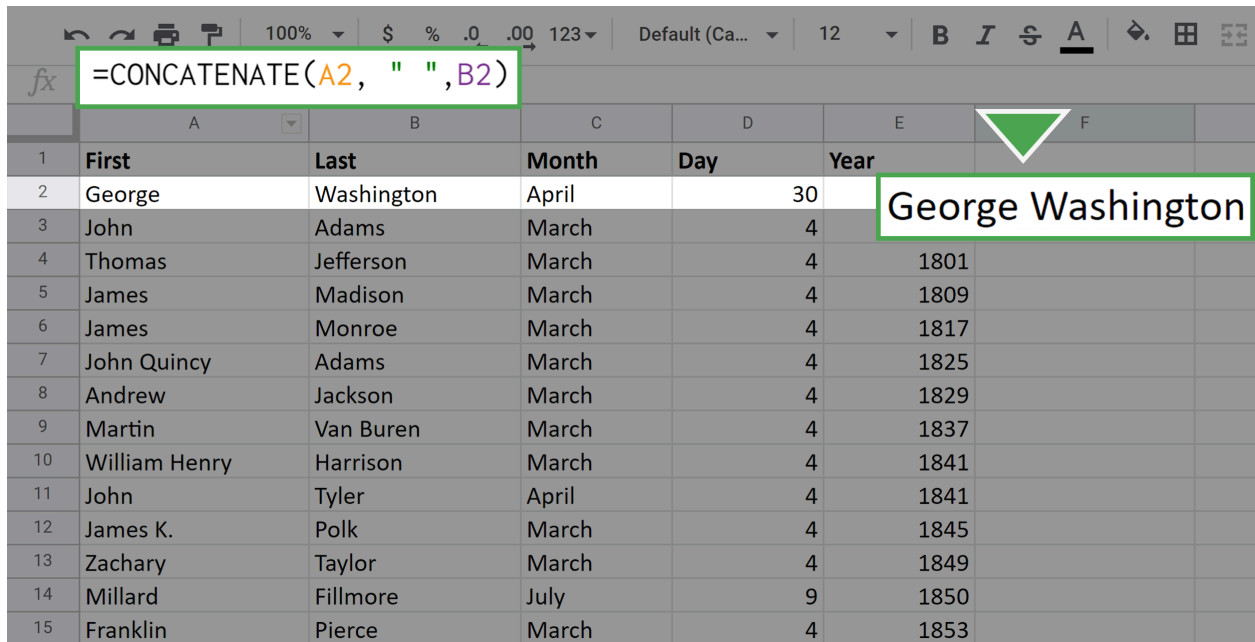
If you want to put the space in between, you need to use the full **CONCATENATE** function, which allows you to combine multiple strings.

2. Click again on the cell **F2**. In the function call, place a space in quotes between **A2** and **B2** separated by commas.



	A	B	C	D	E	F	G	H
1	First	Last	Month	Day	Year			
2	George	Washington	April		30	GeorgeWashington		
3	John	Adams	March		4			
4	Thomas	Jefferson	March		4			
5	James	Madison	March		4	1809		
6	James	Monroe	March		4	1817		
7	John Quincy	Adams	March		4	1825		
8	Andrew	Jackson	March		4	1829		
9	Martin	Van Buren	March		4	1837		
10	William Henry	Harrison	March		4	1841		
11	John	Tyler	April		4	1841		
12	James K.	Polk	March		4	1845		
13	Zachary	Taylor	March		4	1849		
14	Millard	Fillmore	July		9	1850		
15	Franklin	Pierce	March		4	1853		

Once you press enter or return, your screen should appear like this:



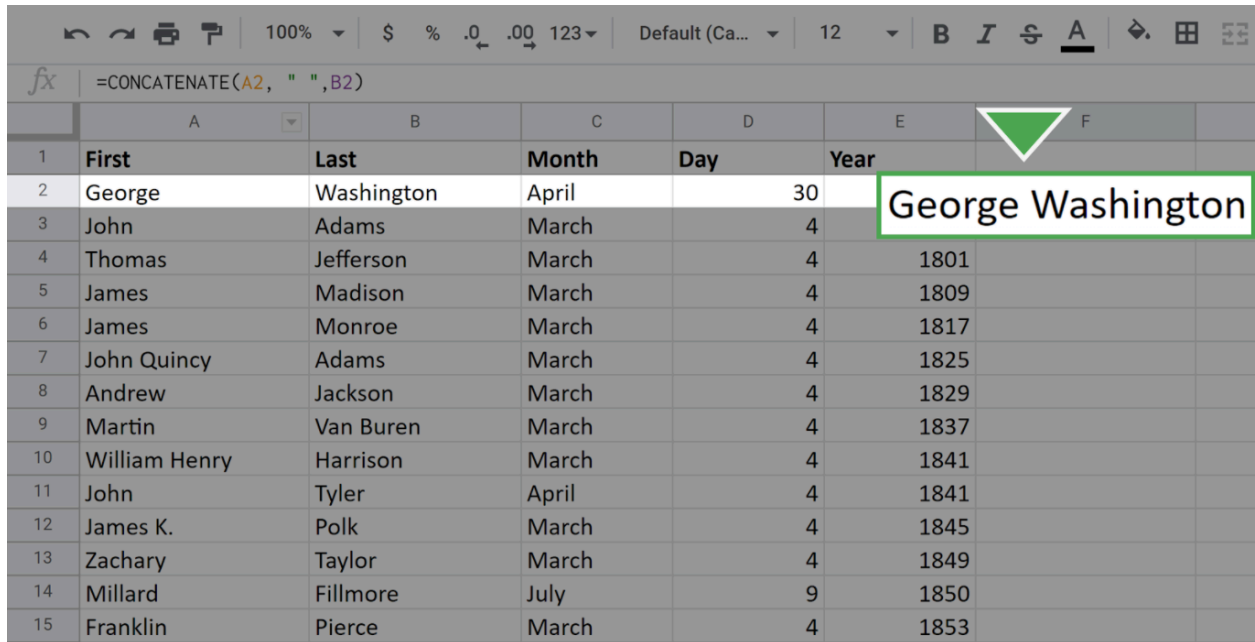
The screenshot shows an Excel spreadsheet with a formula bar at the top displaying `=CONCATENATE(A2, " ", B2)`. The spreadsheet has columns labeled A through F. Column A is labeled 'First', Column B is labeled 'Last', Column C is labeled 'Month', Column D is labeled 'Day', and Column E is labeled 'Year'. Column F is currently empty. The data in the spreadsheet is as follows:

	A	B	C	D	E	F
1	First	Last	Month	Day	Year	
2	George	Washington	April	30		George Washington
3	John	Adams	March	4		
4	Thomas	Jefferson	March	4	1801	
5	James	Madison	March	4	1809	
6	James	Monroe	March	4	1817	
7	John Quincy	Adams	March	4	1825	
8	Andrew	Jackson	March	4	1829	
9	Martin	Van Buren	March	4	1837	
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11	John	Tyler	April	4	1841	
12	James K.	Polk	March	4	1845	
13	Zachary	Taylor	March	4	1849	
14	Millard	Fillmore	July	9	1850	
15	Franklin	Pierce	March	4	1853	

Now there is a space between the first name and the last name.

3. Next, repeat this process for all the remaining cells in **Column F**. Of course, you don't want to do this manually for each cell. (Especially if the dataset were larger, it would be laborious to do this cell-by-cell.) Luckily, you can fill out the data in the column by using your mouse.

4. Click on the cell **F2**. Locate the small square in the lower-right corner of the highlighted boundary of the cell.



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F
1	First	Last	Month	Day	Year	
2	George	Washington	April	30		George Washington
3	John	Adams	March	4		
4	Thomas	Jefferson	March	4	1801	
5	James	Madison	March	4	1809	
6	James	Monroe	March	4	1817	
7	John Quincy	Adams	March	4	1825	
8	Andrew	Jackson	March	4	1829	
9	Martin	Van Buren	March	4	1837	
10	William Henry	Harrison	March	4	1841	
11	John	Tyler	April	4	1841	
12	James K.	Polk	March	4	1845	
13	Zachary	Taylor	March	4	1849	
14	Millard	Fillmore	July	9	1850	
15	Franklin	Pierce	March	4	1853	

5. Click on this square, drag your mouse to the bottom of the column, and release. All the cells in the column should populate with the full name of the appropriate president.

.0 .00 123 Default (Ca... 12			
	C	D	E
	Month	Day	Year
	April	30	George Washington
	March	4	John Adams
	March	4	Thomas Jefferson
	March	4	James Madison
	March	4	James Monroe
	March	4	John Quincy Adams
	March	4	Andrew Jackson
	March	4	Martin Van Buren
	March	4	William Henry Harrison
	March	4	John Tyler
	April	4	James K. Polk
	March	4	Zachary Taylor
	March	4	Millard Fillmore
	July	9	Franklin Pierce
	March	4	

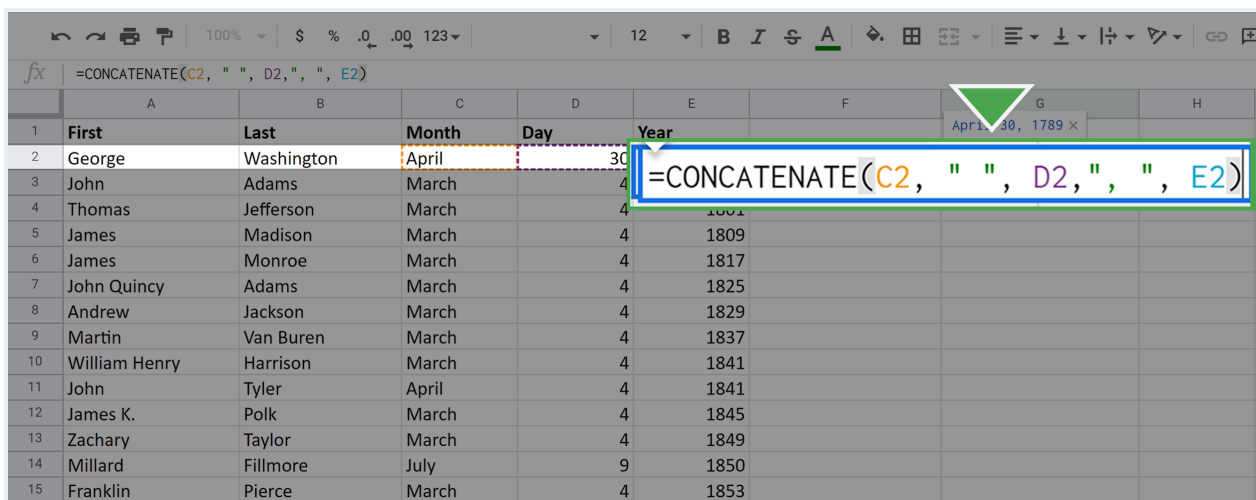
- Note: While it does not happen in this dataset, you may have extra spaces in your result after you `CONCAT`. If you notice you have extra spaces, you can use the `TRIM` function to remove them.

The procedure for combining three pieces of data from different cells is almost identical to what you just did. The only difference is that you include a third cell in the full `CONCATENATE` argument.

Now, combine the month, day, and year into a single data value: **Date**. This will occupy column **G**.

1. Click on the cell where you would like the new data to start. Here, this is cell **G2**.

2. Enter the **CONCAT** command as `=CONCATENATE (C2 , " ", D2 , " , " , E2)`



	A	B	C	D	E	F	G	H
1	First	Last	Month	Day	Year			
2	George	Washington	April	30	1789			
3	John	Adams	March	4	1789			
4	Thomas	Jefferson	March	4	1801			
5	James	Madison	March	4	1809			
6	James	Monroe	March	4	1817			
7	John Quincy	Adams	March	4	1825			
8	Andrew	Jackson	March	4	1829			
9	Martin	Van Buren	March	4	1837			
10	William Henry	Harrison	March	4	1841			
11	John	Tyler	April	4	1841			
12	James K.	Polk	March	4	1845			
13	Zachary	Taylor	March	4	1849			
14	Millard	Fillmore	July	9	1850			
15	Franklin	Pierce	March	4	1853			

Pay particular attention to the extra strings you added between the month and the day, and between the day and the year. This is how you get the spaces and comma in your final result.

			April 30, 1789
			March 4, 1797
	D	E	F
	Day	Year	
	30	1789	George Washington
	4	1797	John Adams
	4	1801	Thomas Jefferson
	4	1809	James Madison
	4	1817	James Monroe
	4	1825	John Quincy Adams
	4	1829	Andrew Jackson
	4	1837	Martin Van Buren
	4	1841	William Henry Harrison
	4	1841	John Tyler
	4	1845	James K. Polk
	4	1849	Zachary Taylor
	9	1850	Millard Fillmore
	4	1853	Franklin Pierce
			March 4, 1853

Congratulations! You've combined data in spreadsheets using the `CONCAT` and `CONCATENATE` functions.

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

In an empty cell in the **CONCAT** function exercise spreadsheet, enter the function **=CONCATENATE (A7, " ", B7)**. What does this function return?

JohnQuincyAdams

John QuincyAdams

John Quincy Adams

#N/A

Correct

The function `=CONCATENATE (A7, " ", B7)` would return "John Quincy Adams." The correct function to use here would be `CONCATENATE`, which you can use to combine strings. Using `CONCAT` with these arguments would resolve this correctly in Microsoft Excel, but return an error in Google Sheets. Going forward, you can use this distinction to write proper spreadsheet functions.

2.

Question 2

In this activity, you used spreadsheet functions to combine data. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- What are some other kinds of data you could combine using `CONCAT` or `CONCATENATE`?
- What happens if a data analyst tries to work with dates or names that haven't been combined with `CONCAT` or `CONCATENATE`?

I can combine any string data, like address, pincode, and country, in one column. or any other kind of data I can concatenate.

If a data analyst can't combine data with `concat` or `concatenate`, then their work will be more lengthy because they have to type manually, and this process takes time. So the `concat`, or `concatenate`, function is very useful and easy to use and save time.

Feedback

Congratulations on completing this hands-on activity! A good response would include how combining data with `CONCAT` and `CONCATENATE` helps you simplify and clean data by combining string data from multiple cells into a single cell.

In order to properly and efficiently analyze data, the data has to be clean and understandable. Data analysts use functions like `CONCAT` to make data easier to work with, which may require combining multiple cells. Coming up, you will learn more spreadsheet functions that will help you analyze data in spreadsheets.